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Fast Approximation Schemes for Two-Stage, Two-Dimensional Bin Packing

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We present an asymptotic fully polynomial time approximation scheme for the two-dimensional generalization of Bin Packing, which requires packing (or cutting) a given set of rectangles from the minimum number of square bins, with the further restriction that packing the rectangles in the bins is done in two stages, as is frequently the case in real-world applications. To the best of our knowledge, this is the first approximation scheme for a nontrivial two-dimensional (and real-world) generalization of a classical one-dimensional packing problem in which rectangles have to be packed in (finite) squares. In addition to the approximability result, we obtain as a byproduct of our analysis an interesting structural result, namely the asymptotic optimality of the lower bound provided by the natural linear programming relaxation of the problem, having one variable for each bin pattern, that can be solved effectively by column generation techniques. Besides its practical relevance, a main reason to concentrate on the two-stage version of the problem is that its study continues to shed light on the general version. In particular, we strongly conjecture that the approximation guarantee of our algorithm for two-dimensional Bin Packing without the two-stage restriction is better than the best known ones. A simplification of the method leads to an approximation scheme for the two-stage version of two-dimensional Strip Packing.

Key words: two-dimensional bin packing; two-dimensional strip packing; two-stage packing; asymptotic fully polynomial time approximation scheme; fractional two-dimensional bin packing
MSC2000 subject classification: Primary: 90C57, 90C05
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1. Introduction. The classical one-dimensional *Bin Packing* (BP) is one of the most widely studied problems in combinatorial optimization. In particular, a huge amount of work has been done to analyze the worst-case performance of heuristic algorithms; see Coffman et al. [7]. The approximability status of BP, namely the existence of *Asymptotic (Fully) Polynomial Time Approximation Schemes* (A(F)PTASs), was essentially settled in the early 80s by Fernandez de la Vega and Lueker [13] and Karmarkar and Karp [18]. In contrast, many open questions exist about the approximability of important two- (and more-)dimensional generalizations of BP, many of which arise in real-world applications.

In this paper, we will consider two-dimensional geometric generalizations of BP, in which rectangles (called also *items* in the following) of specified size have to be packed in other (larger) rectangles (called *bins*). The most studied version of the problem is the one in which a specified edge for each item has to be packed parallel to a specified edge of a bin. Such a requirement is called *orthogonal packing without rotation*. Different results were obtained for the two cases in which, respectively, there is a unique bin (called also *strip*) where one edge is infinitely long, called *Two-Dimensional Strip Packing* (2SP), and there are infinitely many identical bins of finite width and height, called *Two-Dimensional (Finite) Bin Packing* (2BP). For 2SP, a long series of results, starting with Coffman et al. [8] and Baker et al. [1] and ending with the elegant AFPTAS proposed by Kenyon and Rémila [20], settled the approximability of the problem, at least from an asymptotic viewpoint. On the other hand, for 2BP the best known asymptotic approximation guarantee has been 2.125 (Chung et al. [6]) for about fifteen years, until the result of Kenyon and Rémila [20], that easily leads to a $2+\varepsilon$ guarantee (for any $\varepsilon > 0$). Later, motivated by the present work, Caprara [3] proposed an algorithm with guarantee $T_\infty + \varepsilon$ (again for any $\varepsilon > 0$), where $T_\infty = 1.691\ldots$ is the well-known guarantee of the harmonic algorithm for BP (Lee and Lee [22]). Very recently, Bansal and Sviridenko [2] showed that there is no APTAS for 2BP

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unless P = NP. Moreover, Bansal and Sviridenko [2] and Correa and Kenyon [9] show an APTAS for the case in which all the items are *squares*.

The above situation for the case of orthogonal packing without rotation is common to all the two-dimensional variants of bin packing that we are aware of: When the items have to be packed in *finite* bins, there is a huge gap between the approximation guarantee that can be achieved by known methods and the one that is excluded by negative results. The only problem that one may consider as an exception is the classical BP itself, which coincides with the *one-stage* 2BP. In this paper, we take this one step further, deriving an APTAS for the *two-stage* version, called *Two-Dimensional Shelf Bin Packing* (2SBP), which arises in real-world applications. To the best of our knowledge, this is the first approximation scheme for a nontrivial two-dimensional (and real-world) generalization of a classical one-dimensional packing problem in which rectangles have to be packed in (finite) squares. In addition to the approximability result, we obtain as a byproduct of our analysis an interesting structural result, namely the asymptotic optimality of the lower bound provided by the natural linear programming relaxation of the problem, having one variable for each bin pattern, that can be solved effectively by column generation techniques. Besides its practical relevance, a main reason to concentrate on the two-stage version of the problem is that its study continues to shed light on the general version. In particular, we strongly conjecture that the approximation guarantee of our algorithm for 2BP (without the two-stage restriction) is better than the best known one mentioned above, since the optimal solution values with and without imposing the two-stage restriction appear to be fairly close. A simplification of the method leads to an approximation scheme for the two-stage version of 2SP, called *Two-Dimensional Shelf Strip Packing* (2SSP).

Our approximation scheme is unusually structured, being subdivided into several steps. The techniques used in single steps are analogous to those in other approximation schemes in the literature, namely:

- The partitioning of a set of items into subsets such that the ratio between the minimum and maximum size in each subset is bounded by a constant, and the sizes of items in distinct subsets are significantly different, followed by the separate solution of the instances associated with each subset—this is used in a few recent approximation schemes, and to the best of our knowledge was first presented by Skutella and Woeginger [27].
- The well-known *linear grouping* technique of Fernandez de la Vega and Lueker [13], that considers a set of items by decreasing sizes, partitions it into subsets with the same cardinality, and assigns to each item in each subset the largest (or smallest) size in the subset.
- The *geometric grouping* technique of Karmarkar and Karp [18], that is the variant of linear grouping in which the sum of the sizes of the items in each subset (rather than their number) is (approximately) the same.
- The solution of constant-size *Integer Linear Programs* (ILPs) by the method of Lenstra [23], or, in order to improve the running time, the approximate solution of constant-size *Linear Programs* (LPs) by specialized methods (which include the approximate solution of the column generation problem in order to avoid dealing explicitly with all the variables)—the use of these techniques in approximation schemes was introduced by Fernandez de la Vega and Lueker [13] and Karmarkar and Karp [18].

On the other hand, the combination of these techniques into the overall method is far from trivial. Furthermore, many known techniques are applied with a new purpose. For instance, geometric grouping is used in order to round the heights of "small" (thin) items, whereas grouping is used only to round the size of "large" items. For "large" items, we first round the heights and then the widths, achieving a constant number of "large" rectangle sizes. This typically does not work for other geometric problems. Moreover, we solve a number of ILPs in order to jointly pack the "large" and "small" items optimally, whereas for most other schemes, one ILP suffices and "large" and "small" items can be packed separately. The structure of these ILPs appears to be more complex than the one of the ILPs used

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in other approximation schemes. Finally, in order to get fully polynomial running time, we have to approximately solve the LP relaxation of these ILPs through the approximate solution of the associated column generation problem, which by itself is a two-dimensional packing problem—this is the first time that an approximation scheme for a two-dimensional packing problem is used with this scope.

The degree of sophistication of our method, which may be unavoidable, perhaps reflects the intrinsic difficulty in designing approximation schemes (if any exist) for more complex k-stage versions of 2BP with $k \geq 3$.

Finally, letting n denote the number of items and ε the accuracy required, it has to be remarked that our method runs either in $O(n \log n)$ time with a huge multiplicative constant (doubly exponential in $1/\varepsilon$), in case we solve the ILPs optimally, or it has a practically reasonable running time (in the fully polynomial version) but it pays a huge additive constant in the value of the solution produced in order to round the solution of the LP relaxation of the ILPs. Therefore, although it constitutes a considerable improvement with respect to the one in a preliminary version of this paper (Caprara et al. [5]) where the exponent of n in the running time was multiply exponential in $1/\varepsilon$, it is unlikely that our method as it stands will be usable in practice, as is the case for most of the other (A)PTASs we are aware of. On the other hand, from a practical viewpoint, many ideas in our method may inspire the design of (parts of) heuristic algorithms that are effective in practice. Moreover, the asymptotic optimality of the natural LP relaxation of 2SBP, obtained as a byproduct of our analysis, is an interesting theoretical explanation of the empirical behavior observed in practice, and gives further motivation to use such a relaxation to tackle real-world instances of the problem.

Two-stage packing. The *two-stage* name originated from cutting applications in which the items have to be *cut* from the bins. Roughly speaking, two-stage means packing the items in the bins in rows so as to form a sequence of *shelves* (or *levels*); see Figure 1. Formally, according to Gilmore and Gomory [14], two-stage cutting/packing requires that in the final pattern each item can be obtained from the associated bin by at most two stages of cutting. After a first stage in which the bins are cut into *shelves*, the second stage produces *slices* which contain a single item. If an additional stage is allowed in order to separate an item from a waste area, as in Figure 1(a), the problem is called two-stage *with trimming* (also known as the *nonexact case*); otherwise, the problem is called two-stage *without trimming* (the *exact case*), as in Figure 1(b). In this paper we consider the case with trimming since, besides being more frequent in practical applications, it is much closer to the case without stage restrictions, as discussed below.

Staged cutting problems have practical relevance whenever the raw material to be cut has a low cost with respect to the industrial costs involved by the cutting process, as is the case in many wood, paper, or glass cutting applications. Real-world applications involving two-stage cutting problems in the steel industry can be found in De Carvalho and Rodrigues

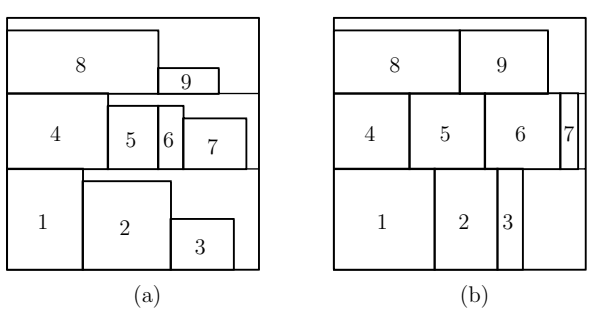

FIGURE 1. Examples of two-staged patterns: Nonexact (a) and exact (b) cases.

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[11, 12], while an industrial three-stage cutting problem was recently considered by Vanderbeck [28].

Note that two-stage packing with trimming is equivalent to packing the items in the bins in shelves, where a *shelf* is a row of items having their bases on a line which is either the base of the bin or the line drawn at the top of the tallest item packed in the shelf preceding the current one in the bin (see again Figure 1(a)).

Two-stage packing may appear highly inefficient if no stage restriction is imposed. However, the asymptotic worst-case ratio between the optimal 2SSP and 2SP solution values is T_∞ $(= 1.691\ldots)$, as shown by Csirik and Woeginger [10]. For the 2BP case, the situation may be even better, since the asymptotic worst-case ratio between the optimal 2SBP and 2BP solution values is not larger than T_∞, as shown by Caprara [3]. However, to the best of our knowledge, this ratio may be as small as $3/2$ provided one chooses the best 2SBP solution between the one for the original instance and the one for the instance in which all items are rotated by 90 degrees (or, equivalently, shelves are formed by packing items one on top of the other, and then shelves are packed in bins one beside the other). In any case, if the ratio were $<T_\infty$, our AFPTAS would be the best known approximation algorithm for 2BP. On the other hand, it is immediate to construct examples for which the optimal solution value is 1 for 2SP and 2BP and arbitrarily close to infinity for the two-stage *exact* cases of the problems.

Finally, note that the approximability status of the exact case is easily settled starting from known results on BP. Specifically, the exact versions of 2SSP and 2SBP are solved by separately packing the items with the same height in the minimum number of shelves (solving a BP for each different height) and then, for 2SBP, by packing these shelves in the minimum number of bins (solving another BP). Since no polynomial-time algorithm for BP can have absolute approximation guarantee better than $3/2$ (unless $\mathrm{P} = \mathrm{NP}$), it is easy to see that the best *asymptotic* approximation guarantee for exact 2SSP and 2SBP cannot be better than $3/2$. This is, on the other hand, easily achieved by forming the shelves using any BP algorithm with absolute guarantee $3/2$ and, for 2SBP, packing the shelves in bins by an APTAS for BP (in this case the guarantee is actually $3/2 + \varepsilon$).

Preliminaries. In the problems we consider, we are given n items, the jth having *width* $w_j \le 1$ and *height* $h_j \le 1$. We also say that the jth item has *size* (w_j, h_j). With an abuse of notation, we will identify items with their indices. Accordingly, the set of items given on input will be denoted by $N = \{1, \ldots, n\}$, and $\mathcal{N}$ will denote the collection of all possible inputs. Given a set $S \subseteq N$, we let $w(S) := \sum_{j \in S} w_j$ and $h(S) := \max_{j \in S} h_j$. A *shelf* is a set $S \subseteq N$ whose *width* $w(S)$ does not exceed 1. The *height* of the shelf is given by $h(S)$. We let $\mathcal{Z}_+$ denote the set of integer nonnegative numbers (including 0).

In 2SSP the items have to be packed in shelves within a strip of width 1 and infinite height so as to minimize the overall height of the shelves. In 2SBP the items have to be packed in shelves and the resulting shelves in identical bins of width and height 1 (i.e., the overall height of the shelves in each bin must not exceed 1) so as to minimize the number of nonempty bins. Clearly, both 2SSP and 2SBP are generalizations of BP, which arises when $h_j = 1$ for all $j \in N$, and therefore APX-hard, so the best approximation one can hope for is an AFPTAS.

Consider one of the above problems, say P. Given a set N of items, let $\mathrm{opt}_{\mathrm{P}}(N)$ denote the value of the optimal solution of P for N. If no confusion arises, we will write $\mathrm{opt}(N)$ for $\mathrm{opt}_{\mathrm{P}}(N)$. An APTAS is an algorithm that receives on input a required *accuracy* $\varepsilon > 0$, runs in time polynomial in the size of the item set N, and produces a solution of value at most $(1 + \varepsilon) \cdot \mathrm{opt}(N) + \delta$ for every N and for some value δ which is independent of N (but may depend on ε). (For a maximization problem, the value of the solution found must be at least $(1 - \varepsilon) \cdot \mathrm{opt}(N) - \delta$.) A *PTAS* is an APTAS for which $\delta = 0$. An *(A)FPTAS* is an (A)PTAS whose running time is polynomial also in $1/\varepsilon$. Throughout the paper, we

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will illustrate (AF)PTASs working with an internal ε, showing that the value of the final solution is at most $1+\alpha\cdot\varepsilon$ times the optimum (plus an additive constant) for some absolute constant α, and omitting the obvious and tedious details concerning the computation of the correct value of ε given the required accuracy on input.

The *Next Fit* (NF) algorithm for BP considers the items in an arbitrary order and packs them in the current bin as long as they fit. When the current item does not fit, the current bin is closed and a new bin is considered, ending the procedure when all items have been packed. The corresponding running time is $O(n)$. For our problems, NF is used to pack both items in shelves and shelves in bins. The following is a well-known fact for BP (stated here in terms of 2SBP):

Lemma 1.1. *Given a set of shelves of total height H (each having height at most 1), these shelves are packed in at most $2H+1$ bins by NF.*

We will also consider the *Fractional* NF (FNF) variant that, given a set of bins already containing some items, proceeds like NF for the remaining items, with the difference that if the current item does not fit in the current bin, it packs the fraction that fits (closing the bin) and then packs the remaining fraction in the next bins. Clearly, such a policy is optimal if the items remaining can be cut.

The *Next Fit Decreasing Height* (NFDH) first proposed and analyzed in Coffman et al. [8] for 2SSP considers the items in *decreasing order of height* and packs them in shelves by a next fit policy, closing the current shelf and starting a new one when the current item does not fit in the shelf. Moreover, for 2SBP, the shelves formed are then packed in bins by NF. The following lemma is due to Coffman et al. [8].

Lemma 1.2. *Given a set of items of total area V (each having height at most 1), these items are packed in shelves of total height at most $2V+1$ by NFDH.*

Combining Lemmas 1.1 and 1.2 yields:

Lemma 1.3. *Given a set of items of total area V (each having height at most 1), these items are packed in at most $4V+3$ bins by NFDH.*

In case the height of some shelves as well as the packing of a subset $S\subseteq N$ of the items in these shelves have already been *fixed*, and the remaining items can be cut vertically, we define the following special version of NFDH, referred to as *Fractional* NFDH (FNFDH). (Note that the height of a shelf may be fixed to a value that is *strictly larger* than the maximum height of the items it contains in S.)

Initially, all shelves are *open* and all items in $R := N\setminus S$ *unpacked*. We consider the tallest open shelf, whose height is (say) h, and the unpacked item $j\in R$ with $h_j\le h$ and maximum. If j completely fits in the shelf (i.e., there is enough horizontal residual space), we pack j in the shelf and iterate. Otherwise, we pack the fraction of j that fits in the shelf, *close* the latter, define a new item that represents the unpacked fraction of j, and iterate. If either all shelves initially available are closed, or the minimum height of an unpacked item is larger than the maximum height of an open shelf, the procedure continues by forming *new* shelves, proceeding as NFDH but cutting the items vertically, and terminating when all items in R are packed.

2. An APTAS for 2SBP.

In this section we illustrate an APTAS for 2SBP, presenting the algorithm in §2.1 and analyzing its performance guarantee in §2.2.

2.1. Description of the algorithm.

The general structure of our APTAS for 2SBP, called $\text{APTAS}_{\text{2SBP}}$, is shown in Figure 2. The internal value ε yields an asymptotic approximation guarantee $1+\alpha\cdot\varepsilon$ for some absolute constant α, as proved in §2.2. Below, we describe in detail the various steps, showing that the running time is polynomial for

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Algorithm APTAS$_{2SBP}$:

(a) **Decomposition:**
- (a.1) Increase the *heights* of a subset of items with small total area so that the item set can be partitioned into sets A_i, $i \in \mathcal{Z}_+$, such that:
  - (i) the ratio between the minimum height in A_i and the maximum height in A_{i+1} is at least $1/\varepsilon$;
  - (ii) the ratio between the maximum and the minimum height in A_i is bounded by a constant;
  - (iii) set A_0 contains all the items with height $\geq \varepsilon$.
- (a.2) Impose the constraint that each shelf in the solution contains items from the same set A_i, $i \in \mathcal{Z}_+$.

(b) **Item Subdivision, Vertical Melting and Grouping:**
For each set A_i, $i \in \mathcal{Z}_+$:
- (b.1) Separate *wide* items (having width $\geq \varepsilon$) from *thin* items (having width $< \varepsilon$).
- (b.2) Allow thin items to be cut *vertically*.
- (b.3) Round the *heights* of the wide items by *linear grouping*.
- (b.4) Round the *widths* of the wide items with the *same height* after Step (b.3) by *linear grouping*.
- (b.5) Round the *heights* of the thin items by *geometric grouping*.

(c) **Definition of Low Shelves:**
For each set A_i, $i \in \mathcal{Z}_+ \backslash \{0\}$:
- (c.1) Pack all items in A_i in shelves of overall minimum height by ILP (1)–(4) combined with FNFDH applied to the thin items.

(d) **Horizontal Melting and Definition of Bins:**
- (d.1) Allow the low shelves found in Step (c) to be cut *horizontally*.
- (d.2) Pack all items in A_0 and all shelves formed in Step (c) in the minimum number of bins by ILP (5)–(9) combined with FNFDH applied to the thin items in A_0 and FNF to the low shelves.

(e) **Size Restoring and Solidification:**
- (e.1) Pack the low shelves that are cut horizontally in additional bins by NF.
- (e.2) Restore the original heights of the thin items, packing those that do not fit in additional bins by FNFDH.
- (e.3) Restore the original widths of the wide items, packing those that do not fit in additional bins by NFDH.
- (e.4) Restore the original heights of the wide items, packing the shelves that do not fit in additional bins by NF.
- (e.5) Pack the thin items that are cut vertically in additional bins by NFDH.

FIGURE 2. General structure of the APTAS for 2SBP.

every fixed ε. Generally, since we modify the widths and heights of the items *several times*, when describing a step, we will denote by w_j and h_j, respectively, the width and height of each item j *before* performing the step.

Decomposition. The general aim of Step (a) is to decompose the original set of items into subsets with the property that the ratio between the maximum and minimum item heights in a subset is bounded by a constant, and there is at least a factor $1/\varepsilon$ between the smallest height in a subset and the largest height in the next subset. The packing is then constructed separately for each subset. Namely, for all subsets except the first one (i.e., the one with the largest heights) items are packed in shelves with the objective of minimizing the overall height of the shelves, whereas the items in the first subset are packed in bins together with the shelves defined for the other subsets with the objective of minimizing the number of bins. On the one hand, if the item heights are significantly different for different subsets, it is nearly optimal to pack only items from the same subset in each shelf. On the other hand, the fact that there is at most a constant factor between the maximum and the minimum height in a subset is essential for the subsequent grouping step, as will be evident from the proofs of Lemmas 2.7, 2.8, and 2.9.

In Step (a.1), we compute the total area of the items $j \in N$ such that $h_j \in \bigcup_{i \in \mathcal{Z}_+} (\varepsilon^{l+it}, \varepsilon^{l+it-1}]$ for $l = 1, \ldots, t := \lceil 1/\varepsilon^2 \rceil$. Then, letting f be the value of l for which this area is minimum, for each item $j \in N$ such that $h_j \in (\varepsilon^{f+it}, \varepsilon^{f+it-1}]$ for some $i \in \mathcal{Z}_+$, we redefine $h_j := \varepsilon^{f+it-1}$.

Then we let $A_i := \{j \in N\colon h_j \in [\varepsilon^{f+it-1}, \varepsilon^{f+(i-1)t}]\}$ for $i \in \mathcal{Z}_+$, noting that $A_0 = \{j \in N\colon h_j \in [\varepsilon^{f-1}, 1]\}$. This guarantees that the ratio between the maximum and the minimum

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height in A_i is at most $\varepsilon^{f+(i-1)t}/\varepsilon^{f+it-1} = 1/\varepsilon^{t-1}$, which is the constant mentioned above, and that the ratio between the minimum height in A_i and the maximum height in A_{i+1} is at least $1/\varepsilon$. Of course, we can avoid considering all sets A_i such that $A_i = \varnothing$. Accordingly, there are $O(n)$ nonempty sets A_i that can be defined in $O(n)$ time. In Step (a.2) we impose the restriction that the items in different sets A_i be packed in different shelves until Step (d). In Step (e), this restriction is removed.

Throughout the description, f will have the meaning defined here and i will be used as the index to identify one of the subsets defined.

Grouping. Grouping is a well-known technique to transform a generic set of items into a set for which there is only a constant number of distinct sizes (widths and heights in our case), without significantly changing the optimal solution value. Having a constant number of sizes allows us to formulate the problem as an ILP with a constant number of variables and constraints. As is customary, we separate items in each set A_i into large and small (wide and thin in our case). For wide items, we first restrict the number of distinct heights to a constant, and then the number of distinct widths for items having the same height to a constant. For thin items, that we allow to be cut vertically, we simply restrict the number of distinct heights to a constant.

Let $C_i := \{j \in A_i\colon w_j \geq \varepsilon\}$ and $B_i := \{j \in A_i\colon w_j < \varepsilon\}$ be, respectively, the set of *wide* and *thin* items in A_i ($i \in \mathcal{I}_+$). For our purposes, in order to prove the results in §4.1, it is convenient to define grouping so as to *reduce* the size of the items.

In Steps (b.3) and (b.4), *linear grouping* is the classical grouping proposed in Fernandez de la Vega and Lueker [13]. Namely, in Step (b.3), let $h_1 \geq h_2 \geq \cdots \geq h_{l_i}$ be the heights of the items in C_i after sorting. If $l_i < 2/\varepsilon^{t+1}$, we leave these heights unchanged, defining for convenience $g_i := l_i$. Otherwise, letting $k_i := \lfloor l_i\varepsilon^{t+1}\rfloor$, we form $g_i := \lceil l_i/k_i\rceil$ groups of k_i consecutive items, starting from the k_i tallest ones, and letting the last group contain $l_i - (g_i - 1)k_i$ items. We redefine the heights of the items in each group as the height of the *last* (*lowest*) item in the group. After this step, the number g_i of different heights in C_i is upper bounded by $3/\varepsilon^{t+1} + 1 = O(1/\varepsilon^{t+1})$. Indeed, if $l_i < 2/\varepsilon^{t+1}$, we have $g_i = l_i$. Otherwise, letting $a \geq 2$ be such that $a \leq l_i\varepsilon^{t+1} < a+1$, the number of heights is

$$\begin{aligned} g_i = \left\lceil \frac{l_i}{\lfloor l_i\varepsilon^{t+1}\rfloor} \right\rceil &\leq \frac{l_i}{\lfloor l_i\varepsilon^{t+1}\rfloor} + 1 \leq \frac{l_i}{l_i\varepsilon^{t+1} - 1} + 1 \\ &\leq \frac{1}{\varepsilon^{t+1}}\left(\frac{a+1}{a-1}\right) + 1 \leq 3/\varepsilon^{t+1} + 1, \end{aligned}$$

since $(a+1)/(a-1) \leq 3$ for $a \geq 2$.

In Step (b.4), if $l_i < (g_i+1)/\varepsilon^{t+1}$, we leave the widths of all items in C_i unchanged. Otherwise, we consider *separately* the sets of items in C_i with the same height h_j after Step (b.3), for $j = 1, \ldots, g_i$, each denoted by D_{ij}. For each set D_{ij}, let $w_1 \geq w_2 \geq \cdots \geq w_{m_{ij}}$ be the widths of the items after sorting. We form $d_{ij} := \lceil m_{ij}/p_i\rceil$ groups of $p_i := \lfloor l_i\varepsilon^{t+1}/g_i\rfloor$ consecutive items starting from the p_i widest ones and letting the last group contain $m_{ij} - (d_{ij} - 1)p_i$ items (possibly forming only one group). We redefine the width of the items in each group as the width of the *last* (*thinnest*) item in the group. After this step, the number of different widths in C_i, and therefore of different sizes, is upper bounded by $g_i(g_i+2)/\varepsilon^{t+1} + g_i = O(1/\varepsilon^{3(t+1)})$. This trivially holds if $l_i < (g_i+1)/\varepsilon^{t+1}$. Otherwise, letting $a \geq g_i + 1$ be such that $a \leq l_i\varepsilon^{t+1} < a+1$, the overall number of widths in C_i is

$$\begin{aligned} \sum_{j=1}^{g_i} d_{ij} &\leq \sum_{j=1}^{g_i} \frac{m_{ij}}{l_i\varepsilon^{t+1}/g_i - 1} + 1 = g_i\left(\frac{l_i}{l_i\varepsilon^{t+1} - g_i} + 1\right) \\ &\leq \frac{g_i}{\varepsilon^{t+1}}\left(\frac{a+1}{a-g_i}\right) + g_i \leq g_i(g_i+2)/\varepsilon^{t+1} + g_i. \end{aligned}$$

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According to the above discussion, for each $i \in \mathcal{I}_+$, the number s_i of distinct item sizes in each set C_i after Steps (b.3) and (b.4) is $O(1/\varepsilon^{3(t+1)})$.

In Step (b.5) we group the heights of the thin items (whose widths are unchanged afterwards) according to the *geometric grouping* of Karmarkar and Karp [18]. To this aim, let B_i be the set of thin items considered and $h_1 \geq h_2 \geq \cdots \geq h_{r_i}$ be the heights of the items in B_i after sorting. We form groups with consecutive items, starting from the first (tallest) one, so that the overall *width* of the items in each group adds up to *exactly* $\varepsilon^t w(B_i)$, with the possible exception of the last group, for which the overall width of the items is $(1 - \lfloor 1/\varepsilon^t \rfloor \varepsilon^t) w(B_i)$ (if this value is not 0). Note that such a subdivision is always possible since we allow thin items to be cut vertically, and therefore we may insert only a vertical fraction of an item in a group. As for linear grouping, we redefine the heights of the items in each group as the height of the *last* (*lowest*) item in the group. The overall number of heights for the items in B_i after this step is $q_i := \lceil 1/\varepsilon^t \rceil = O(1/\varepsilon^t)$. We can represent each group by a single *equivalent item* that may be cut vertically.

Solution by constant-size ILPs. In Steps (c) and (d) we separately pack the items in each set A_i ($i \in \mathcal{I}_+$) by solving suitable ILPs with a constant number of variables. This is possible since the number of distinct item sizes is bounded by a constant after Step (b). In particular, in Step (c) we consider each set A_i, $i \in \mathcal{I}_+ \backslash \{0\}$, and solve the associated 2SSP instance, optimally packing the items in so-called *low* shelves so as to minimize the overall height of the shelves. In Step (d), we consider the set A_0 along with the shelves produced in Step (c) and optimally solve the associated 2SBP instance; i.e., we pack the items in A_0 in shelves and the shelves in bins, together with the low shelves (that may be cut horizontally), so as to minimize the number of bins used. As a side remark, note that, differently from the (A)PTASs for the one-dimensional BP and other similar problems, it would be *wrong* to pack the items in A_0 in the minimum number of bins and *only later* pack the low shelves in these bins, since there may be alternative ways of packing the items in A_0 in the minimum number of bins with significantly different total height of the resulting shelves, and therefore residual space for the low shelves.

For each set A_i ($i \in \mathcal{I}_+$), recall that the number s_i of different sizes for the wide items in C_i is bounded by a constant, letting (w_l, h_l) be the lth size and n_l the corresponding number of items ($l = 1, \ldots, s_i$), and that the number of different heights for the wide items is $O(1/\varepsilon^{t+1})$. Moreover, recall that $q_i = O(1/\varepsilon^t)$ is the number of groups of thin items, letting $(\widetilde{w}_j, \tilde{h}_j)$ ($j = 1, \ldots, q_i$) denote the size of the equivalent item. For convenience, we assume $\tilde{h}_1 > \tilde{h}_2 > \cdots > \tilde{h}_{q_i}$ (*strict* inequality). This is allowed since we can replace possible groups with the same height by a unique group of items (hence not all $\widetilde{w}_j$ are necessarily equal).

For Step (c), consider a generic set A_i ($i \in \mathcal{I}_+ \backslash \{0\}$). We let a *shelf configuration* S be defined by a *height* h_S, chosen among the $O(1/\varepsilon^{t+1})$ possible heights of the wide and thin items, and by a number n_{lS} of wide items for $l = 1, \ldots, s_i$, such that $n_{lS} = 0$ if $h_l > h_S$ and $w_S := \sum_{l=1}^{s_i} n_{lS} w_l \leq 1$. Since the width of each wide item is at least ε, the latter requirement imposes a bound of $O((1/\varepsilon + 1)^{s_i})$ on the number of possibilities for the n_{lS} values. Let $\mathcal{S}$ denote the collection of all possible shelf configurations, noting that $|\mathcal{S}|$ is constant for fixed ε although doubly exponential in $1/\varepsilon$. Note that if a shelf associated with configuration $S \in \mathcal{S}$ is formed in the solution, the residual horizontal space $\overline{w}_S := 1 - w_S$ can be used to pack thin items whose height does not exceed h_S. For $j = 1, \ldots, q_i$ let $\mathcal{S}_j := \{S \in \mathcal{S}\colon h_S \geq \tilde{h}_j\}$, noting that $\mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \cdots \subseteq \mathcal{S}_{q_i} \subseteq \mathcal{S}$.

In the following ILP, the variables x_S, $S \in \mathcal{S}$, represent the number of shelves associated with configuration S in the solution:

$$\min \sum_{S \in \mathcal{S}} h_S x_S, \tag{1}$$

$$\sum_{S \in \mathcal{S}} n_{lS} x_S \geq n_l, \quad l = 1, \ldots, s_i, \tag{2}$$

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$$\sum_{S \in \mathcal{S}_j} \overline{w}_S x_S \geq \sum_{k=1}^{j} \widetilde{w}_k, \quad j = 1, \ldots, q_i, \tag{3}$$

$$x_S \geq 0 \text{ integer}, \quad S \in \mathcal{S}. \tag{4}$$

Objective function (1) requires the minimization of the overall height of the shelves; constraints (2) ensure that all the wide items are packed; and constraints (3) guarantee that there is enough space for the thin items in the shelves, noting that these items can be cut vertically and that thin items of height $\tilde{h}_j$ can be packed only in shelves associated with configurations in $\mathcal{S}_j$. Since the number of variables is constant, this ILP can be solved in polynomial time by the algorithm of Lenstra [23]. (Note also that the number of constraints is constant, namely $O(1/\varepsilon^{3(t+1)})$.) Given an optimal solution of (1)–(4), the thin items are packed in the shelves selected by FNFDH without having to introduce additional shelves.

Step (d) is an extension of Step (c). Shelf configurations are defined in exactly the same way (considering now the items in A_0). Moreover, a *bin configuration* B is defined by specifying a number n_{SB} of shelf configurations S for each $S \in \mathcal{S}$, where $h_B := \sum_{S \in \mathcal{S}} n_{SB} h_S \leq 1$. Since $h_S \geq \varepsilon^{f-1} \geq \varepsilon^{t-1}$ for $S \in \mathcal{S}$ (the height of all items in A_0 being at least ε^{f-1}), letting $\mathcal{B}$ denote the collection of all bin configurations, we have $|\mathcal{B}| = O((1/\varepsilon^{t-1} + 1)^{|\mathcal{S}|})$; i.e., $|B|$ is constant for fixed ε, although triply exponential in $1/\varepsilon$. If a bin associated with configuration $B \in \mathcal{B}$ is packed in the solution, there is a residual horizontal space for the thin items in A_0 in the shelves associated with the shelf configurations in the bin (as for the previous ILP) as well as a residual vertical space $\bar{h}_B := 1 - h_B$ for the low shelves formed in Step (c) (that can be cut horizontally). In order to pack all these shelves, we must simply be sure that the overall residual vertical space in the bins is not smaller than the overall height of these shelves, say H, which is the sum of the solution values of ILP (1)–(4) over all $i \in \mathcal{I}_+ \backslash \{0\}$. Accordingly, letting y_B denote the number of bins associated with configuration B in the solution, we have the following ILP:

$$\min \sum_{B \in \mathcal{B}} y_B, \tag{5}$$

$$\sum_{B \in \mathcal{B}} \left(\sum_{S \in \mathcal{S}} n_{lS} n_{SB} \right) y_B \geq n_l, \quad l = 1, \ldots, s_0, \tag{6}$$

$$\sum_{B \in \mathcal{B}} \left(\sum_{S \in \mathcal{S}_j} \overline{w}_S n_{SB} \right) y_B \geq \sum_{k=1}^{j} \widetilde{w}_k, \quad j = 1, \ldots, q_0, \tag{7}$$

$$\sum_{B \in \mathcal{B}} \bar{h}_B y_B \geq H, \tag{8}$$

$$y_B \geq 0 \quad \text{integer}, \quad B \in \mathcal{B}. \tag{9}$$

As already mentioned, constraint (8) is essential; i.e., solving the above ILP without (8) and then packing the low shelves in the bins by FNF would not necessarily lead to an optimal solution. Again, the number of variables is constant and the ILP can be solved in polynomial time. (The number of constraints is $O(1/\varepsilon^{3(t+1)})$ as for ILP (1)–(4).) Given an optimal solution of (5)–(9), the thin items in A_0 are packed in the shelves selected by FNFDH, and the low shelves are packed in the bins selected by FNF, without having to introduce additional shelves and bins.

The optimal solution of ILP (1)–(4) for each $i \in \mathcal{I}_+ \backslash \{0\}$ (followed by FNFDH) and the subsequent solution of (5)–(9) (followed by FNFDH and FNF) solves optimally the problem after the redefinition of widths and heights, provided the thin items can be cut vertically and the low shelves can be cut horizontally.

As discussed in §3, in order to achieve an APTAS, it would be enough to solve the LP relaxations of the above ILPs and round up the corresponding solutions.

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Size restoring and solidification. The purpose of Step (e) is to define a feasible solution for the original instance starting from the solution found in Step (d), in which low shelves may be cut horizontally, thin items may be cut vertically, and the widths and heights of the items are in general smaller than those in the instance after Step (a).

In the solution found in Step (d), at most one low shelf per bin is cut. These shelves, whose height does not exceed ε^f, are packed in Step (e.1) in additional bins by NF.

In Steps (e.2)–(e.4) we transform the solution for the instance with sizes modified (reduced) according to Steps (b.2)–(b.4) into a solution for the instance with the original sizes, at the cost of using extra bins. As is customary, we will replace an item (or shelf) j by some other item (or shelf) k ensuring that the original size of k is not larger than the modified size of j. It is easy to check that the replacements below have this property. The remaining largest items (or shelves) will then be packed in additional bins by some greedy algorithm. As already mentioned, we will remove the requirement imposed by Step (a.2) and allow items from different sets A_i to be packed in the same shelf. In order to describe this size-restoring part, we adopt the same notation used in the description of Step (b).

In Step (e.2), for each set A_i ($i \in \mathcal{I}_+$), for $j = 1, \ldots, q_i - 1$, we use the space occupied by the thin items in group j in order to pack the items in group $j + 1$ with their original heights (before Step (b.5)). The remaining items in the first group for each $i \in \mathcal{I}_+$ are then packed in additional bins by FNFDH (recalling that these items can be cut vertically).

In Step (e.3), for each set A_i ($i \in \mathcal{I}_+$), we consider the wide items with the same height h_j in each set D_{ij} ($j = 1, \ldots, g_i$), sorted by decreasing widths, and, if the widths of these items were changed in Step (b.4), for $k = 1, \ldots, m_{ij} - p_i$, we use the space occupied by the kth item in order to pack the $(k + p_i)$th one with its original width. (If $m_{ij} < p_i$ we do not use the space occupied by the m_{ij} items in D_{ij}.) The widest items remaining for each $i \in \mathcal{I}_+$ and $j = 1, \ldots, g_i$ are then packed in additional bins by NFDH.

In Step (e.4), we consider the, say, l_i shelves associated with the items in each set A_i ($i \in \mathcal{I}_+$), letting $h_1 \geq h_2 \geq \cdots \geq h_{l_i}$ be the corresponding heights. If the heights of these items were changed in Step (b.3), for $j = 1, \ldots, l_i$ we use the space occupied by the jth shelf in order to pack the $(j + k_i)$th shelf after having restored the original heights of the items in the shelf. The tallest remaining shelves for each $i \in \mathcal{I}_+$ are then packed in additional bins by NF.

Finally, in Step (e.5), we consider thin items in each set A_i ($i \in \mathcal{I}_+$) that are cut vertically in the solution. Recalling how FNFDH works, we know that there is at most one such item per shelf in the solution. We use NFDH to pack these items in additional bins.

Overall, the running time of $\text{APTAS}_{\text{2SBP}}$ is polynomial for fixed ε. We already noted that both the number of variables and constraints in the ILPs for the solution of Steps (c) and (d) is constant and that implies that the time required to solve them is polynomial in the size of the coefficients, all of which are $O(n)$. Accordingly, the bottleneck of the asymptotic running time is the sorting of the items, which is required in several places, leading to the following.

Lemma 2.1. *$\text{APTAS}_{\text{2SBP}}$ runs in $O(n \log n)$ time for every fixed ε.*

2.2. Proof of approximation guarantee. In order to prove that our algorithm produces a near-optimal solution, we analyze each step in Figure 2. Note that Steps (b) and (d.1) correspond to a *relaxation* of the problem, therefore the optimal solution value cannot increase after performing these steps. Moreover, Step (d.2) finds an optimal solution of the problem resulting after the previous steps; hence it cannot be responsible for an increase in the optimal solution value. On the other hand, Step (a) defines a new instance for which the solution value is worse than before. Moreover, Step (c) defines the shelves for the items in $\bigcup_{i \in \mathcal{I}_+ \setminus \{0\}} A_i$ so as to guarantee that the overall height of the shelves is minimized—note that this is not necessarily optimal for 2SBP. Finally, Step (e) performs an adjustment of the (infeasible) solution, increasing the number of bins.

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We analyze the steps that increase the solution value in order of appearance in the algorithm. In the analysis, we will often implicitly assume $\varepsilon \leq \frac{1}{2}$.

Lemma 2.2. *Letting z and z' denote the optimal solution values before and after Step* (a.1), *respectively, we have $z' \leq (1+4\varepsilon)z+3$.*

Proof. Let V denote the overall area of the items and note that $z \geq V$. By an obvious average argument, for the value of f considered, the overall area of the items whose heights are rounded up is not larger than V/t. Moreover, the increase in the height, and therefore in the area, of these items in Step (a.1) is by a factor at most $1/\varepsilon$. Therefore, considering an optimal solution of value z for the original instance and packing the items with size modified after Step (a.1) in additional bins by NFDH yields a solution of value at most $z+4V/(t\varepsilon)+3 \leq (1+4\varepsilon)z+3$ by Lemma 1.3, noting that $1/(t\varepsilon) \leq \varepsilon$ by the definition of t. □

Lemma 2.3. *Letting z and z' denote the optimal solution values before and after Step* (a.2), *respectively, we have $z' \leq (1+4\varepsilon)z+1$.*

Proof. Recall that Step (a.2) requires that the items in each set A_i be packed in distinct shelves. Consider an optimal solution for the instance before decomposition, letting H denote the overall height of the corresponding shelves and noting that $z \geq H$. Consider a shelf with items $S \subseteq N$ and let $P := \{i \in \mathcal{I}_+ \colon S \cap A_i \neq \varnothing\}$. If $|P| = 1$, this shelf is also feasible after decomposition. Otherwise, letting h_S be the height of the shelf and k the minimum index in P, we may leave the items in $S \cap A_k$ in the shelf as well as define $|P|-1$ new shelves, one for each set $S \cap A_i$, $i \in P \backslash \{k\}$. Since the height of each item in A_{k+i+1} is not larger than ε^{1+it} times the height of each item in A_k ($i \in \mathcal{I}_+$), the overall height of these new shelves does not exceed

$$\sum_{i \in \mathcal{I}_+ \colon k+i+1 \in P} \varepsilon^{1+it} h_S \leq \sum_{i \in \mathcal{I}_+} \varepsilon^{1+it} h_S = \varepsilon \sum_{i \in \mathcal{I}_+} (\varepsilon^t)^i h_S = \frac{\varepsilon}{1-\varepsilon^t} h_S \leq 2\varepsilon h_S.$$

Therefore, the overall height of the new shelves after performing the above for all shelves in the original solution does not exceed $2\varepsilon H$; i.e., the new shelves can be packed in at most $4\varepsilon H+1$ additional bins by NF by Lemma 1.1. □

Lemma 2.4. *If the low shelves can be cut horizontally* (*Step* (d.1)), *then it is optimal to define these shelves so that their overall height is minimized* (*Step* (c)).

Proof. Obvious. □

Lemma 2.5. *Letting z and z' denote the values of the solutions constructed before and after Step* (e.1), *we have $z' \leq (1+2\varepsilon)z+1$.*

Proof. Note that z is the number of bins in the solution after Step (d). Recalling how FNF works, we know that at most z low shelves are cut horizontally in this solution. Their overall height is at most $\varepsilon^f z$ since the height of each low shelf is at most ε^f. Hence, these shelves are packed in at most $2\varepsilon^f z+1 \leq 2\varepsilon z+1$ bins by NF by Lemma 1.1. □

For convenience, in the proofs of Lemmas 2.6–2.8 below we use the same notation as in the description of Step (b) in §2.1.

Lemma 2.6. *Letting z and z' denote the values of the solutions constructed before and after Step* (e.2), *we have $z' \leq (1+4\varepsilon)z+3$.*

Proof. Let $W_i := w(B_i)$ denote the overall width of the thin items in A_i and H_i the overall height of shelves with items in A_i in the solution before restoring the heights of the thin items, noting that $z \geq \sum_{i \in \mathcal{I}_+} H_i$. We have $H_i \geq W_i \varepsilon^{f+it-1}$ since each shelf can contain thin items for a total width of at most 1 and the height of all these items is at least ε^{f+it-1}. The total width of the remaining items in A_i (that in Step (e.2) are packed in additional bins)

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is $\varepsilon^t W_i$; therefore their overall area is at most $\varepsilon^t W_i \cdot \varepsilon^{f+(i-1)t} \leq \varepsilon H_i$. Hence, the overall area of the remaining items (considering all sets A_i) is at most $\sum_{i \in \mathcal{I}_+} \varepsilon H_i = \varepsilon z$, and these items are packed in no more than $4\varepsilon z + 3$ bins by FNFDH by Lemma 1.3. □

Lemma 2.7. *Letting z and z' denote the values of the solutions constructed before and after Step* (e.3), *we have $z' \leq (1+4\varepsilon)z + 3$.*

Proof. For each set A_i, let H_i be the overall height of the corresponding shelves in the solution before restoring the widths of the wide items, noting that $z \geq \sum_{i \in \mathcal{I}_+} H_i$. We have $H_i \geq l_i \varepsilon^{f+it}$ since at most $1/\varepsilon$ wide items can be in a shelf, and the height of each item is at least ε^{f+it-1}. Considering the items in set D_{ij}, we have at most p_i remaining items that in Step (e.3) are packed in additional bins. Then the total height of the remaining items in A_i, and therefore their total area, is at most $g_i p_i \varepsilon^{f+(i-1)t} \leq \varepsilon H_i$ by the definition of l_i, g_i, and p_i. Hence, the overall area of the remaining items (considering all sets A_i) is at most $\sum_{i \in \mathcal{I}_+} \varepsilon H_i = \varepsilon z$, and these items are packed by NFDH in no more than $4\varepsilon z + 3$ bins by Lemma 1.3. □

Lemma 2.8. *Letting z and z' denote the values of the solutions constructed before and after Step* (e.4), *we have $z' \leq (1+2\varepsilon)z + 1$.*

Proof. For each set A_i, let H_i be the overall height of the corresponding shelves in the solution before restoring the heights of the wide items, noting that $z \geq \sum_{i \in \mathcal{I}_+} H_i$. As in the proof of Lemma 2.7, we have $H_i \geq l_i \varepsilon^{f+it}$. Moreover, there are at most k_i shelves that are packed in additional bins in Step (e.4), whose total height is at most $k_i \varepsilon^{f+(i-1)t} \leq \varepsilon H_i$ by the definition of l_i and k_i. Accordingly, the overall height of the remaining shelves (considering all sets A_i) is at most $\sum_{i \in \mathcal{I}_+} \varepsilon H_i \leq \varepsilon z$, and they are packed by NF in at most $2\varepsilon z + 1$ additional bins by Lemma 1.1. □

Lemma 2.9. *Letting z and z' denote the values of the solutions constructed before and after Step* (e.5), *we have $z' \leq (1+4\varepsilon)z + 3$.*

Proof. Let H be the overall height of the shelves in the solution, noting that $z \geq H$. For each shelf S, recalling how FNFDH works, we know that at most one thin item j of width $w_j \leq \varepsilon$ and height $h_j \leq h_S$, and therefore area at most εh_S, is cut vertically. Accordingly, the overall area of the thin items cut is at most εH, and these items are packed in at most $4\varepsilon H + 3$ bins by NFDH by Lemma 1.3. □

According to Lemmas 2.2–2.9, the final solution produced by $\text{APTAS}_{\text{2SBP}}$ for an instance I has value at most

$$(1+2\varepsilon)^2(1+4\varepsilon)^5 \text{opt}(N) + 15 \leq (1+\alpha \cdot \varepsilon)\text{opt}(N) + 15,$$

for some absolute constant α depending on the maximum possible value of ε. For instance, if $\varepsilon \leq \frac{1}{2}$, we have $\alpha = 1942$. Hence, the following theorem follows from Lemmas 2.1–2.9.

Theorem 2.1. *$\text{APTAS}_{\text{2SBP}}$ is an APTAS for* 2SBP *that runs in $O(n \log n)$ time for fixed ε.*

3. Achieving fully polynomial running time. Clearly, the constant hidden behind the $O(n \log n)$ mentioned in Theorem 2.1 is huge. On the other hand, we next show how to modify the method so as to achieve fully polynomial running time at the cost of having an additive constant hidden behind the asymptotic approximation guarantee which is (singly) exponential in $1/\varepsilon$.

Consider ILPs (1)–(4) and (5)–(9) and note first of all that they have at most $2|A_i| + 1 = O(n)$ constraints, as the number of sizes cannot be larger than the number of items. Since the number of variables is huge, in analogy to the AFPTAS for BP by Karmarkar and Karp [18], we can restrict ourselves to the solution of the associated LP relaxations by avoiding explicitly considering all variables, using *column generation* in order to introduce

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variables only if needed, and then rounding up the resulting LP solution. Moreover, since a near-optimal solution of the LP relaxations is sufficient, we can solve the column generation problem approximately (see Karmarkar and Karp [18], Plotkin et al. [26], Grigoriadis et al. [15]). In §3.1 we discuss this issue, whereas in §3.2 we describe and analyze the resulting AFPTAS.

3.1. Approximate solution of the LP relaxations. Although each of the methods illustrated in Karmarkar and Karp [18], Plotkin et al. [26], and Grigoriadis et al. [15] can be applied to our case, we will adopt the one in Grigoriadis et al. [15]. It is much faster than the one of Karmarkar and Karp [18], avoids the use of the ellipsoid method, and applies more naturally to our case with respect to the one of Plotkin et al. [26], since the time complexity of the latter depends on the so-called *width* of the convex set P (see below), which is in principle unbounded in our case. It is possible to reduce the width to a value which is single exponential in $1/\varepsilon$, which yields a fully polynomial running time for an improved version of the method of Plotkin et al. [26], for which the time complexity increases only logarithmically with the width. However, all these technicalities can be skipped using the method of Grigoriadis et al. [15].

The method of Grigoriadis et al. [15]. Consider an LP of the form

$$r^* := \min\{c^T x\colon Mx \geq e, x \geq 0\}, \tag{10}$$

where c is a nonnegative cost vector, M is a nonnegative $m \times n$ matrix, and for convenience, e and 0 denote respectively, a unit and a null vector of appropriate dimension. This problem can be solved approximately as follows by the method of Grigoriadis et al. [15] that, given a tentative solution value r, works with the optimization problem

$$\lambda^* := \max\{\lambda\colon Mx \geq \lambda e, x \in P\}, \tag{11}$$

where $P := \{x\colon c^T x = r, x \geq 0\}$ (actually the method is much more general—here we will restrict attention to what is needed for our purposes). In particular, the method needs a subroutine to approximately solve the following *block problem*:

$$\mu^* := \max\{v^T Mx\colon x \in P\} \tag{12}$$

for a given nonnegative vector v. Since P is a simplex, the optimum of this problem will be attained at a vertex of the simplex, in which only one of the components is positive; i.e., an optimal solution of (12) is defined by $x_{j^*} = r/c_{j^*}$ and $x_j = 0$ $(j \neq j^*)$, where

$$j^* := \arg\max_{1 \leq j \leq n} \frac{\sum_{i=1}^m v_i m_{ij}}{c_j}. \tag{13}$$

Since the maximum in (13) is larger than one if and only if the reduced cost of x_{j^*} with respect to v is negative, the block problem turns out to be an optimization version of the column generation problem for LP (10).

The main result of Grigoriadis et al. [15] is the following:

Theorem 3.1. *For every required accuracy ε, a solution x of (11) such that $Mx \geq (1-\varepsilon)\lambda^* e$ can be found by $O(m/\varepsilon^2 + m\log m)$ calls to a procedure that finds a solution x of (12) such that $v^T Mx \geq (1-\varepsilon/6)\mu^*$. If the running time of this latter procedure is $O(T)$, the running time per iteration is $O(m\log(m/\varepsilon) + T)$. The number of nonzero components in the final solution x of (11) is $O(m/\varepsilon^2 + m\log m)$.*

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Corollary 3.1. *For every required accuracy $\varepsilon \le \frac{1}{2}$, a solution x of (10) with $O(m/\varepsilon^2 + m\log m)$ nonzero components and whose value is at most $(1+6\varepsilon)r^*$ can be found in time*

$$O(\log(r_{\max}/(\varepsilon r_{\min}))\cdot(m/\varepsilon^2 + m\log m)\cdot(m\log(m/\varepsilon)+T)),$$

provided $r^* \in [r_{\max}, r_{\min}]$ and there is a procedure that finds a solution x of (12) such that $v^T Mx \ge (1-\varepsilon/6)\mu^*$ and runs in time $O(T)$.

Proof. Let x be the solution returned by the method of Theorem 3.1 for the tentative value r. If $Mx < (1-\varepsilon)e$, then $\lambda^* < 1$; i.e., there is no solution of value r for LP (10). Otherwise $\tilde{x} := (1+2\varepsilon)x$ is a feasible solution of LP (10) whose value is $(1+2\varepsilon)r$. Accordingly, by performing binary search over the range of possible values for r^*, say $[r_{\min}, r_{\max}]$, with discretization $\delta := \varepsilon r_{\min}$, one ends up with a value r such that there is no solution of value $\le r-\delta$; i.e., $r^* > r-\delta \ge (1-\varepsilon)r$, along with a solution $\tilde{x}$ of value $(1+2\varepsilon)r \le (1+2\varepsilon)r^*/(1-\varepsilon) \le (1+2\varepsilon)^2 r^* \le (1+6\varepsilon)r^*$. The number of iterations is $O(\log(r_{\max}/\delta)) = O(\log(r_{\max}/(\varepsilon r_{\min})))$. In combination with the statement of Theorem 3.1, this proves the claim. □

Note finally that a *basic* solution of (10) has at most m nonzero components. Given a solution x with $O(m/\varepsilon^2 + m\log m)$ nonzero components, a basic solution of the same value can be found in a standard way by repeating the following operation $O(m/\varepsilon^2 + m\log m)$ times: select $m+1$ positive components, letting M' be the associated $(m+1)\times m$ submatrix of M, find a nonzero solution of $M'z = 0$ and use it to decrease by one the number of nonzero components in x without affecting feasibility and the objective value. The resulting complexity is $O(\mathcal{M}\cdot(m/\varepsilon^2 + m\log m))$, where $\mathcal{M}$ is the time to invert an $m\times m$ matrix.

Application to our case. In order to apply the method of Grigoriadis et al. [15] to our case, we should first normalize the LPs considered so that all coefficients in the right-hand side are equal to 1. This leads to the following LP relaxations of (1)–(4) and (5)–(9), respectively:

$$\begin{aligned}
\min\ & \sum_{S\in\mathcal{S}} h_S z_S \\
& \sum_{S\in\mathcal{S}} \frac{n_{lS}}{n_l} z_S \ge 1, \quad l = 1,\ldots,s_i, \\
& \sum_{S\in\mathcal{S}_j} \frac{\overline{w}_S}{\sum_{k=1}^{j}\widetilde{w}_k} z_S \ge 1, \quad j = 1,\ldots,q_i, \\
& z_S \ge 0, \quad S\in\mathcal{S},
\end{aligned} \tag{14}$$

and

$$\begin{aligned}
\min\ & \sum_{B\in\mathcal{B}} y_B \\
& \sum_{B\in\mathcal{B}} \frac{\sum_{S\in\mathcal{S}} n_{lS} n_{SB}}{n_l} y_B \ge 1, \quad l = 1,\ldots,s_0, \\
& \sum_{B\in\mathcal{B}} \frac{\sum_{S\in\mathcal{S}_j} \overline{w}_S n_{SB}}{\sum_{k=1}^{j}\widetilde{w}_k} y_B \ge 1, \quad j = 1,\ldots,q_0, \\
& \sum_{B\in\mathcal{B}} \frac{\bar{h}_B}{H} y_B \ge 1, \\
& y_B \ge 0, \quad B\in\mathcal{B}.
\end{aligned} \tag{15}$$

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We will consider only the sets A_i for which the total width of the items is larger than 1, since otherwise the items in A_i can be trivially packed in a unique shelf and no LP is needed.

Note that $m = O(\min\{|A_i|, 1/\varepsilon^{3(t+1)}\})$, recalling that groups of thin items with the same height are actually considered as a single group in these models. We will need $m = O(|A_i|)$ in the running time analysis and $m = O(1/\varepsilon^{3(t+1)})$ in the performance guarantee proof, as the additive error will turn out to be $O(mh)$, where h is the maximum height in A_i.

Column generation for LP (14). The dual of (14) reads:

$$\begin{aligned} \max \quad & \sum_{l=1}^{s_i} \pi_l + \sum_{j=1}^{q_i} \sigma_j, \\ & \sum_{l=1}^{s_i} \frac{n_{lS}}{n_l} \pi_l + \sum_{j=j_S}^{q_i} \frac{\overline{w}_S}{\sum_{k=1}^{j} \widetilde{w}_k} \sigma_j \le h_S, \quad S \in \mathscr{S}, \\ & \pi_l, \sigma_j \ge 0, \quad l = 1, \ldots, s_i;\ j = 1, \ldots, q_i, \end{aligned} \tag{16}$$

where $j_S := \min\{j\colon \tilde{h}_j \le h_S\}$ $(S \in \mathscr{S})$ denotes the index of the maximum height of a thin item that fits in a shelf associated with configuration S. Given nonnegative vectors π, σ, the block problem (12) associated with (14), recalling its equivalent expression (13), can be stated as:

$$\max_{S \in \mathscr{S}} \frac{1}{h_S} \left(\sum_{l=1}^{s_i} \frac{n_{lS}}{n_l} \pi_l + \sum_{j=j_S}^{q_i} \frac{\overline{w}_S}{\sum_{k=1}^{j} \widetilde{w}_k} \sigma_j \right). \tag{17}$$

We can rewrite the problem as $\max_{S \in \mathscr{S}}(p_S/h_S)$, where the *shelf profit* p_S of each shelf configuration $S \in \mathscr{S}$ is given by:

$$p_S := \sum_{l=1}^{s_i} \frac{n_{lS}}{n_l} \pi_l + \sum_{j=j_S}^{q_i} \frac{\overline{w}_S}{\sum_{k=1}^{j} \widetilde{w}_k} \sigma_j. \tag{18}$$

Let S^* denote an optimal solution and p_{S^*} the corresponding profit. Given objects with profits and weights, each available in unlimited number of copies, and a knapsack of unit capacity, the *Unbounded Knapsack Problem* (UKP) calls for packing a certain number of copies of each object in the knapsack so as to maximize the profit packed. The natural ILP formulation of the problem has an integer variable for each object, representing the number of copies of the object that are packed. Note that we use the term *object* to keep the distinction from the *items* in the 2SBP instance.

For h_S (and therefore also j_S) fixed, recalling $\overline{w}_S := 1 - \sum_{l=1}^{s_i} n_{lS} w_l$, the problem (17) of finding a shelf with maximum profit is an UKP with integer variables n_{lS}, each associated with an object of profit π_l/n_l and weight w_l, with the additional constraint $n_{lS} = 0$ if $h_l > h_S$, and an additional *continuous* object having profit $\sum_{j=j_S}^{q_i}(\sigma_j/\sum_{k=1}^{j} \widetilde{w}_k)$ and weight 1, for which any fraction may be inserted in the knapsack. An easy way of solving (17) approximately is then to try all the $O(|A_i|)$ possibilities for h_S, applying an (absolute) FPTAS for the corresponding UKP, that is easily extended to handle the continuous object. Adapting the FPTAS for the ordinary Knapsack Problem by Lawler [21] would yield an overall complexity which is quadratic in $|A_i|$, considering all possibilities for h_S. In the appendix, we illustrate a simple adaptation of this method whose complexity is only linear in $|A_i|$, proving the following lemma:

Lemma 3.1. *For every required accuracy ε, a solution S of the block problem* (17) *associated with LP* (14) *such that $p_S \ge (1-\varepsilon)p_{S^*}$, can be found in $O(|A_i|/\varepsilon^2)$ time.*

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Column generation for LP (15). The situation for (15) is similar, although slightly more involved since the block problem itself turns out to be a two-dimensional (two-stage) packing problem. The dual of (15) reads:

$$\begin{aligned} \max \quad & \sum_{l=1}^{s_0} \pi_l + \sum_{j=1}^{q_0} \sigma_j + \tau, \\ & \sum_{l=1}^{s_0} \frac{\sum_{S\in\mathscr{S}} n_{lS} n_{SB}}{n_l} \pi_l + \sum_{j=1}^{q_0} \frac{\sum_{S\in\mathscr{S}_j} \overline{w}_S n_{SB}}{\sum_{k=1}^{j} \widetilde{w}_k} \sigma_j + \frac{\bar{h}_B}{H} \tau \le 1, \quad B \in \mathscr{B}, \\ & \pi_l, \sigma_j, \tau \ge 0, \quad l = 1, \ldots, s_0;\ j = 1, \ldots, q_0. \end{aligned} \tag{19}$$

Given nonnegative π, σ, τ, the block problem (12) associated with (15), recalling (13), can be stated as:

$$\max_{B\in\mathscr{B}} \sum_{l=1}^{s_0} \frac{\sum_{S\in\mathscr{S}} n_{lS} n_{SB}}{n_l} \pi_l + \sum_{j=1}^{q_0} \frac{\sum_{S\in\mathscr{S}_j} \overline{w}_S n_{SB}}{\sum_{k=1}^{j} \widetilde{w}_k} \sigma_j + \frac{\bar{h}_B}{H} \tau. \tag{20}$$

Recalling the definition of p_S in (18), we can rewrite the problem as $\max_{B\in\mathscr{B}} p_B$, where the *bin profit* p_B of each bin configuration $B \in \mathscr{B}$ is given by:

$$p_B := \sum_{S\in\mathscr{S}} p_S n_{SB} + \frac{\bar{h}_B}{H} \tau. \tag{21}$$

Let B^* denote an optimal solution and let p_{B^*} be the corresponding profit. As before, since $\bar{h}_B := 1 - \sum_{S\in\mathscr{S}} n_{SB} h_S$, the problem (20) of finding a bin with maximum profit is an UKP whose objects are associated with integer variables n_{SB}, each having profit p_S and weight h_S, and one additional continuous object having profit τ/H and weight 1.

Lemma 3.2. *For every required accuracy ε, a solution B of the block problem (20) associated with LP (15) such that $p_B \ge (1-\varepsilon)p_{B^*}$ can be found in $O(|A_0|/\varepsilon^2)$ time.*

Proof. Clearly, instead of computing p_S for all shelf configurations and then solving an UKP with $|\mathscr{S}|$ variables, we can limit the number of variables in the UKP to $O(|A_0|)$ by considering, for each possible value h of h_S, only the shelf configuration $S^*(h)$ such that $p_{S^*(h)} = \max\{p_S\colon S \in \mathscr{S}, h_S \le h\}$. In fact, we can simply find a configuration $S(h)$ with $h_{S(h)} \le h$ and $p_{S(h)} \ge (1-\varepsilon)p_{S^*(h)}$. The method to approximately solve (17) illustrated in the appendix finds such configurations for *all* h. We can then apply the same FPTAS to solve the UKP associated with (20). The resulting bin configuration B satisfies $p_B \ge (1-\varepsilon)^2 p_{B^*} \ge (1-2\varepsilon)p_{B^*}$. □

Note that the solution method above shows how to easily approximate the Two-Stage, Two-Dimensional UKP in which items (rectangles) have arbitrary profits (not necessarily equal to the area) and are available in unlimited numbers of copies. It is important to note that the latter condition is essential, since the regular (bounded) version is known to admit no FPTAS unless P = NP, even if all item heights are equal to $\frac{1}{2}$ and all item profits are equal to the area (see Caprara et al. [4]).

Overall method. The approximate solution of LPs (14) and (15) proceeds by performing binary search with discretization $\delta := \varepsilon^{f+it}$, according to Corollary 3.1 and the following lemma:

Lemma 3.3. *For both LPs (14) and (15) the optimal solution value is contained in the interval $[\varepsilon^{f+it-1}, |A_i|\varepsilon^{f+(i-1)t}]$.*

Proof. The upper bound is associated with the trivial (integer) solution that packs each item in a separate shelf (and each shelf in a separate bin for set A_0). The lower bound holds since the total width of the items in $|A_i|$ is at least 1, observing that the optimal value of the two LPs is not smaller than the overall area of the associated items. □

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For each tentative solution value, we use the procedure of Theorem 3.1 solving the block problems by the methods of Lemmas 3.1 and 3.2.

LEMMA 3.4. *For every required accuracy $\varepsilon \le \frac{1}{2}$, a solution of both LPs* (14) *and* (15) *with $O(1/\varepsilon^{3t+5}\log(1/\varepsilon))$ nonzero components and whose value is at most $(1+6\varepsilon)$ times the optimum can be found in time*

$$O((\log|A_i| + 1/\varepsilon^2\log(1/\varepsilon)) \cdot (|A_i|/\varepsilon^2\log(1/\varepsilon)) \cdot (|A_i|/\varepsilon^2\log(1/\varepsilon))).$$

PROOF. The number of nonzero components is $O(m/\varepsilon^2 + m\log m) = O(1/\varepsilon^{3t+5} \cdot \log(1/\varepsilon))$, by Corollary 3.1. The proof of the approximation guarantee is immediate from Corollary 3.1, considering Lemmas 3.1, 3.2, and 3.3 and the choice of the binary search step δ. (Note that an internal accuracy of $\varepsilon/6$ is required for the procedures of Lemmas 3.1 and 3.2.) As to running time, for $i \in \mathcal{I}_+$ the time complexity $O(T)$ in the statement of Corollary 3.1 is $O(|A_i|/\varepsilon^2)$, and the number of binary search iterations is bounded by $O(\log(|A_i|/\varepsilon^t)) = O(\log|A_i| + 1/\varepsilon^2\log(1/\varepsilon))$. Therefore, since $m = O(\min\{|A_i|, 1/\varepsilon^{3(t+1)}\})$, the running time given in the statement follows by noting that $m/\varepsilon^2 + m\log m = O(|A_i|/\varepsilon^2 + |A_i|\log(1/\varepsilon^{3(t+1)})) = O(|A_i|/\varepsilon^2\log(1/\varepsilon))$, recalling that $t = O(1/\varepsilon^2)$, and $m\log(m/\varepsilon) = O(|A_i|/\varepsilon^2\log(1/\varepsilon))$. □

3.2. The AFPTAS. Let $\text{AFPTAS}_{\text{2SBP}}$ denote the algorithm obtained from $\text{APTAS}_{\text{2SBP}}$ by replacing the solution of ILPs (1)–(4) in Step (c) and the solution of ILPs (5)–(9) in Step (d) by the approximate solution of their LP relaxations, proceeding as illustrated in §3.1, followed by the rounding up of the LP solutions. Namely, the approximate solutions of LPs (14) and (15) are transformed in integer solutions by simply rounding up each component to the nearest integer value.

THEOREM 3.2. *$\text{AFPTAS}_{\text{2SBP}}$ is an AFPTAS for* 2SBP *that runs in $O(n^2/\varepsilon^4\log^2(1/\varepsilon) \cdot (\log n + 1/\varepsilon^2\log(1/\varepsilon)))$ time.*

PROOF. Noting that the asymptotic running time is dominated by the time required by the approximate solution of the LPs, the running time follows from Lemma 3.4, being

$$\begin{aligned} &O\left(\sum_{i\in\mathcal{I}_+}(\log|A_i| + 1/\varepsilon^2\log(1/\varepsilon)) \cdot (|A_i|/\varepsilon^2\log(1/\varepsilon)) \cdot (|A_i|/\varepsilon^2\log(1/\varepsilon))\right) \\ &\quad = O\left(\sum_{i\in\mathcal{I}_+}|A_i|^2/\varepsilon^4\log^2(1/\varepsilon) \cdot (\log|A_i| + 1/\varepsilon^2\log(1/\varepsilon))\right) \\ &\quad = O(n^2/\varepsilon^4\log^2(1/\varepsilon)(\log n + 1/\varepsilon^2\log(1/\varepsilon))). \end{aligned}$$

As far as the approximation guarantee is concerned, by Corollary 3.1, the number of nonzero components in each of the LP solutions is $O(m/\varepsilon^2 + m\log m)$. Since for all LPs we have $m = O(1/\varepsilon^{3(t+1)})$, the overall increase in the value of each LP (14) due to rounding is $O((m/\varepsilon^2 + m\log m)\varepsilon^{f+(i-1)t}) = O(\varepsilon^{it}(1/\varepsilon^{4t+4}\log(1/\varepsilon)))$, recalling $f \ge 1$. Summing over all $i \in \mathcal{I}_+\backslash\{0\}$ yields an additive constant H' that must be added to H in LP (15), which is $O(1/\varepsilon^{3t+4}\log(1/\varepsilon))$. It is easy to check that this addition increases the optimal value of the latter LP by at most H'. Finally, the increase in the final number of bins by rounding up the solution of this LP is $O(1/\varepsilon^{3t+5}\log(1/\varepsilon))$. Then, the approximation guarantee follows from the one for the APTAS in §2. □

Of course, the additive constant could be decreased by preprocessing the final LP solutions so as to make them basic, as outlined in §3.1, increasing the time complexity. In this case, the increase in the final solution value due to rounding would be $O(1/\varepsilon^{3(t+1)})$.

The effect of solving the LP relaxations and then rounding is then to increase the heuristic solution value by a constant, which does not affect the asymptotic approximation guarantee, although the additive constant is now (singly) exponential in $1/\varepsilon$ whereas before it was independent of ε.

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4. Further implications and extensions. In this section we discuss issues related to the approximation schemes of the previous sections. An interesting structural result that comes as a byproduct is the asymptotic optimality of a well-known LP relaxation of 2SBP. Moreover, we discuss the approximation guarantee of our method for the general 2BP and show how to modify it so as to get an A(F)PTAS for 2SSP.

4.1. The asymptotic optimality of the natural LP relaxation of 2SBP**.** The natural ILP formulation of 2SBP, presented in Gilmore and Gomory [14], can be thought of as being obtained from (5)–(9) by considering all items as "tall" and "wide." This relaxation is generally presented by grouping together the items of the same size, letting (w_i, h_i) be the ith size, n_i the corresponding number of items, and letting m be the total number of distinct sizes, with $M := \{1, \ldots, m\}$. A shelf configuration $S \in \mathcal{S}$ is now defined by a height h_S equal to h_i for some $i \in M$, and by m values n_{iS} $(i \in M)$ such that $\sum_{i\in M} w_i n_{iS} \le 1$ and $n_{iS} = 0$ if $h_i > h_S$. (Note that in this case we can restrict attention to the case in which $h_S = \max\{h_i\colon i \in M, n_{iS} > 0\}$.) As before, a bin configuration $B \in \mathcal{B}$ is defined by $|\mathcal{S}|$ values n_{SB} $(S \in \mathcal{S})$ such that $\sum_{S\in\mathcal{S}} h_S n_{SB} \le 1$. Again, it is essential not to impose the upper bound of n_i on the number of items of size i contained in a bin configuration in order to have fast (approximation) algorithms for the column generation part. The LP relaxation of the resulting ILP formulation reads:

$$\begin{aligned} \min \quad & \sum_{B\in\mathcal{B}} y_B \\ & \sum_{B\in\mathcal{B}} \left(\sum_{S\in\mathcal{S}} n_{iS} n_{SB} \right) y_B \ge n_i, \quad i \in M, \\ & y_B \ge 0, \quad B \in \mathcal{B}. \end{aligned} \tag{22}$$

In particular, it is easy to see that the exact solution of the column generation problem can be carried out in $O(mc)$ time by dynamic programming, where c is the bin height and width obtained after having scaled all widths and heights to integers (see Gilmore and Gomory [14], Hifi [17]). By a fundamental result of Grötschel et al. [16], Padberg and Rao [25], and Karp and Papadimitriou [19], this implies that LP (22) can be solved to optimality in pseudopolynomial time, which turns out to be small in practice for real-world instances. Alternatively, the method of Grigoriadis et al. [15] described above can be applied to obtain a near-optimal solution.

Our aim is to show that the optimal value of 2SBP and of *Fractional* 2SBP (F2SBP), defined by (22), asymptotically coincide. Since the number of constraints in (22) is $O(m) = O(n)$, straightforward rounding up of a solution may increase the value by a large amount. Therefore, for our purposes, we will go once more through the analysis of algorithm $\text{AFPTAS}_{\text{2SBP}}$, taking into account what happens to the value of (22) in Steps (a) and (b).

Theorem 4.1.

$$\lim_{z\to\infty} \sup \left\{ \frac{\text{opt}_{\text{2SBP}}(N)}{\text{opt}_{\text{F2SBP}}(N)} \colon N \in \mathcal{N}, \text{opt}_{\text{2SBP}}(N) \ge z \right\} = 1.$$

Proof. Fix any positive value $\varepsilon > 0$ and consider $\text{AFPTAS}_{\text{2SBP}}$ with this internal value. The analysis of $\text{AFPTAS}_{\text{2SBP}}$ in the previous sections shows that the relationship between the final solution value produced by the method, say z^{H}, and the optimal solution value of LP (15), say z^{LP}, is $z^{\text{H}} \le (1 + \alpha \cdot \varepsilon) z^{\text{LP}} + O(1/\varepsilon^{3(t+1)})$ (in case a basic solution is found) for some absolute constant α. In the following, we show that $z^{\text{LP}} \le (1 + 4\varepsilon)^2 \text{opt}_{\text{F2SBP}}(N) + 4$, yielding

$$\begin{aligned} \text{opt}_{\text{F2SBP}}(N) &\le \text{opt}_{\text{2SBP}}(N) \le z^{\text{H}} \le (1 + \alpha \cdot \varepsilon) z^{\text{LP}} + O(1/\varepsilon^{3(t+1)}) \\ &\le (1 + \alpha \cdot \varepsilon)(1 + 4\varepsilon)^2 \text{opt}_{\text{F2SBP}}(N) + O(1/\varepsilon^{3(t+1)}), \end{aligned}$$

that yields the proof by taking the limit for $\varepsilon \to 0$.

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Note that LP (15), where H is the sum of the solution values of (14) for $i \in \mathcal{I}_+ \backslash \{0\}$, is the counterpart of (22) in which:

(i) some heights are rounded up, according to Step (a.2);

(ii) shelves are formed by considering separately the items in each set A_i, according to Step (a.2);

(iii) thin items may be cut vertically and heights and widths are rounded (reduced), according to Step (b), and shelves with low items may be cut horizontally, according to Step (d.1).

Note in particular that Steps (a.2) and (d.1) lead to a decomposition of the resulting LP into LPs (14) and (15). Of course (iii) reduces the value of the resulting LP; therefore, it is sufficient to prove that $\mathrm{opt}_{\mathrm{F2SBP}}(N)$ is close to the value of the LP obtained after (i) and (ii). To this aim, we next show that Lemmas 2.2 and 2.3 also hold considering the optimal solutions of (22) before and after performing the associated steps. This will prove $z^{\mathrm{LP}} \leq (1+4\varepsilon)^2 \mathrm{opt}_{\mathrm{F2SBP}}(N) + 4$ and conclude the proof.

In order to extend Lemma 2.2, it is sufficient to note that $\mathrm{opt}_{\mathrm{F2SBP}}(N) \geq V$, where V is the overall area of the items, and that the solution produced by NFDH is also feasible for F2SBP.

As to Lemma 2.3, consider an optimal solution of (22) before Step (a.2). A feasible solution of F2SBP can be thought of as the definition of a set of bins, each containing a set of shelves and with an associated *value* $v \in (0, 1]$ (equal to 1 in the 2SBP case). We say that all the shelves in the bin have *value* v, and each item is *covered* for a fraction v every time it appears in one of these shelves, the overall fraction of the item that is covered in the overall solution being at least 1. Since $\sum_{S \in \mathcal{S}} n_{SB} h_S \leq 1$ for $B \in \mathcal{B}$, the value z of this solution satisfies

$$z \geq \sum_{B \in \mathcal{B}} y_B \sum_{S \in \mathcal{S}} n_{SB} h_S.$$

Starting from this solution, we next define a solution of (22) that is feasible after Step (a.2).

To simplify the exposition, for each bin configuration B such that $y_B > 1$, we imagine $\lceil y_B \rceil$ copies of variable y_B, the first $\lceil y_B \rceil - 1$ with value 1 and the $\lceil y_B \rceil$th with value $y_B - (\lceil y_B \rceil - 1)$. Consider a bin configuration $B \in \mathcal{B}$ such that $y_B > 0$ and shelf configuration $S \in \mathcal{S}$ such that $n_{SB} > 0$ and $|P| > 2$, where $P := \{i \in \mathcal{I}_+ \colon n_{iS} > 0 \text{ for some size in } A_i\}$, letting k be the minimum index in P. As in the proof of Lemma 2.2, we define $|P| - 1$ new shelves, one for each index in $P \backslash \{k\}$, and the overall height of these new shelves does not exceed $2\varepsilon h_S$. Moreover, all these shelves have value y_B. Let us call *area* the product of the height times the value of each shelf. Note that the overall area of the new shelves obtained from all $y_B > 0$ is at most $2\varepsilon \sum_{B \in \mathcal{B}} y_B \sum_{S \in \mathcal{S}} n_{SB} h_S$. We pack these shelves in separate bins by the following *next fit decreasing value* method, that considers the shelves according to decreasing values and packs them in a bin as long as they fit, i.e., as long as the sum of the heights of the shelves does not exceed 1, closing the bin and considering a new one afterwards. The value associated with a bin constructed in this way is set to the maximum value of a shelf it contains.

Packing shelves in bins by decreasing value and then setting the value of each bin to the maximum value of the shelf it contains is perfectly analogous to packing items in shelves by decreasing height and then setting the height of each shelf to the maximum height of an item it contains, as is done by NFDH. Hence, by Lemma 1.2, the total value of the bins constructed this way does not exceed $2V + 1$, where V is the total area of the new shelves; i.e., it is at most

$$4\varepsilon \sum_{B \in \mathcal{B}} y_B \sum_{S \in \mathcal{S}} n_{SB} h_S + 1 \leq 4\varepsilon z + 1.$$

The corresponding change in the solution of (22), to get a solution feasible after Step (a.2) of value at most $(1+4\varepsilon)z + 1$, is straightforward. $\square$

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4.2. Better approximation for 2BP**?** Of course, our approximation schemes are also approximation algorithms for the classical 2BP without level (i.e., two-stage) restrictions. The performance guarantee is in this case given by the (asymptotic) worst-case ratio between the optimal 2BP and 2SBP solutions. As already mentioned, if the horizontal and vertical directions are fixed in advance, this worst-case ratio is equal to T_∞ and an approximation guarantee arbitrarily close to this value can be achieved by the algorithm of Caprara [3] which is much simpler and faster than the approximation schemes of this paper. On the other hand, consider the optimal 2BP solution, of value $\mathrm{opt}_{2\mathrm{BP}}(N)$, and the best of the two 2SBP solutions obtained by interchanging the roles of widths and heights (i.e., also considering *vertical* shelves), of value, say, $\mathrm{opt}_{2\mathrm{SBP}}(N)$. Letting

$$\gamma := \limsup_{z\to\infty} \left\{ \frac{\mathrm{opt}_{2\mathrm{SBP}}(N)}{\mathrm{opt}_{2\mathrm{BP}}(N)} \colon N \in \mathcal{N}, \mathrm{opt}_{2\mathrm{BP}}(N) \geq z \right\},$$

the following is implied by Theorem 2.1:

Corollary 4.1. *Algorithm* $APTAS_{2\mathrm{SBP}}$ *is a polynomial-time asymptotic* $(\gamma+\varepsilon)$*-approximation algorithm for* 2BP *for any* $\varepsilon > 0$.

As anticipated in the introduction, we state the following conjecture, that would imply that $\mathrm{APTAS}_{2\mathrm{SBP}}$ is the approximation algorithm for 2BP with the best approximation guarantee known at present:

Conjecture 4.1. $\gamma < T_\infty (= 1.691\ldots)$.

As mentioned in the introduction, the best lower bound we are aware of on the correct value of γ is 3/2.

4.3. Extension to 2SSP**.** The algorithm described in §2.1, the corresponding analysis in §2.2, and the extension in §3 can easily be modified to get an APTAS and an AFPTAS for 2SSP, the case in which the items must be packed in shelves within a unique bin of infinite height.

To handle the 2SSP case, it is sufficient to scale all item heights so that they are all $\leq \varepsilon^t$; i.e., $A_0 = \varnothing$, skipping Steps (d) and (e.1) and packing the items in additional shelves rather than in additional bins in Step (e). We call the resulting algorithms $\mathrm{APTAS}_{2\mathrm{SSP}}$ and $\mathrm{AFPTAS}_{2\mathrm{SSP}}$. The proof of the following theorems is analogous to the proof of Theorems 2.1 and 3.2.

Theorem 4.2. $APTAS_{2\mathrm{SSP}}$ *is an APTAS for* 2SSP *that runs in* $O(n \log n)$ *time for fixed* ε.

Theorem 4.3. $AFPTAS_{2\mathrm{SSP}}$ *is an AFPTAS for* 2SSP *that runs in* $O(n^2/\varepsilon^4 \log^2(1/\varepsilon) \cdot (\log n + 1/\varepsilon^2 \log(1/\varepsilon)))$ *time.*

Concluding remarks. This paper settles the approximability of the two-stage 2BP, showing that this simplified version is now understood to a large extent. The natural extensions of this research would be the derivation of A(F)PTASs for three-stage 2BP, increasing by one the number of stages, and three-stage three-dimensional BP, increasing by one both the number of dimensions and the number of stages. Note that two-stage three-dimensional BP coincides with 2SBP. Both generalizations appear far from trivial, if at all possible.

For the three-stage two-dimensional case, shelves are obtained by putting vertical "stacks" of items (rather than simply items) one beside each other, where each stack is obtained by putting items one on top of the other along their width edge, the width of the stack being given by the maximum width of its items and the height of the stack by the sum of the heights of its items. The main obstacle against the generalization of our method appears to be the fact that shelf heights in general do not coincide with item heights, and therefore both the decomposition and height grouping steps in our methods do not work any more.

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We also note that the results in Caprara [3] imply that the approximation guarantee of our approximation scheme for three-stage 2BP is $T_\infty + \varepsilon$, as for general 2BP.

For the three-stage three-dimensional case, calling width, height and depth the three dimensions of each item, shelves are formed by a two-stage two-dimensional packing of some items constructed by considering only their widths and depths (where now depths are playing the role of heights in the two-stage packing). The height of a shelf is again the maximum height of an item it contains, and shelves are stacked in bins. (Note that this problem is more general than 2SBP but *not* than three-stage 2BP.) In this case, it is still fine to decompose the item set according to heights, and further according to depths. On the other hand, whereas grouping heights, depths, and widths for wide items poses no problem, it is absolutely unclear how to group heights and/or depths for the thin items, nor how to proceed without this grouping (in fact, grouping the heights of the thin items appears to be a key step of our APTAS).

Nevertheless, we feel that succeeding for three-stage 2BP (or for three-stage three-dimensional BP) seems likely to lead to approximation schemes for d-stage 2BP (or d-stage d-dimensional BP) for any fixed d, making considerable progress with respect to the current state of the art.

Appendix: Proof of Lemma 3.1. For simplicity, we first illustrate the method for the case $\sigma_j = 0$ for $j = 1, \ldots, q_i$; i.e., the profit of the continuous objects is null. Note that the sizes of the wide items in the 2SBP instance correspond to the objects in the UKP instance. In addition to the standard UKP data, each object l also has a height h_l ($l = 1, \ldots, s_i$). Our description assumes some familiarity with knapsack algorithms; for a detailed description of these algorithms we refer the reader to Martello and Toth [24].

Consider the objects sorted by increasing heights, say $h_1 \le h_2 \le \cdots \le h_{s_i}$. For $l = 1, \ldots, s_i$, we want a near-optimal solution of the instance with the first l objects only. We let $U_l := \max_{k=1}^{l} \pi_k / w_k$ denote the optimal value of the continuous relaxation of UKP for the first l objects, and z_l the corresponding optimal integer solution value. It is known and easy to check that $z_l \le U_l \le 2z_l$. Our method keeps a lower bound $\underline{U}_l$ on U_l such that $U_l/2 < \underline{U}_l \le U_l$ and, for each l, distinguishes if the profit π_l of the last object is $> \varepsilon \underline{U}_l$ or not, where ε is an internal parameter proportional to the accuracy required. In the first case, the object is considered *large* and its profit is rounded down to the nearest multiple of $\delta_l := \varepsilon^2 \underline{U}_l$, letting $\tilde{\pi}_l := \lfloor \pi_l / \varepsilon^2 \underline{U}_l \rfloor$. Otherwise, the object is considered *small*, and we keep track of the small object $g_l \in \{1, \ldots, l\}$ for which π_{g_l}/w_{g_l} is maximum.

We also maintain a table F such that $F_{p,l}$ is the minimum capacity required to achieve a profit of $p\delta_l$ (after rounding down) by packing large objects in $1, \ldots, l$, for $p = 0, \ldots, \lfloor 2/\varepsilon^2 \rfloor$. (Actually, it turns out that $F_{p,l}$ is a *lower bound* on this minimum capacity, as will be clear from the proof of Lemma 3.1 below.) The approximate solution for the first l objects is given by the best solution obtained by considering all the solutions corresponding to the entries in the lth column of F whose value is ≤ 1, and completing them with the addition of as many copies of object g_l as there fit in the residual knapsack space.

Initially, we set $F_{p,0} := \infty$ for $p = 0, \ldots, \lfloor 2/\varepsilon^2 \rfloor$ and $\underline{U}_1 := U_1 = \pi_1/w_1$. Computation of the lth column of F from the $(l-1)$th one proceeds as follows. First of all, we compute the ratio between U_l and $\underline{U}_{l-1}$. If this ratio is smaller than 2, we simply let $\underline{U}_l := \underline{U}_{l-1}$ and initialize $F_{p,l} := F_{p,l-1}$ for $i = 0, \ldots, \lfloor 2/\varepsilon^2 \rfloor$. Otherwise, we compute the integer t such that $U_l/2 < 2^t \underline{U}_{l-1} \le U_l$, let $\underline{U}_l := 2^t \underline{U}_{l-1}$, and keep only the entries of the $(l-1)$th column whose index is a multiple of 2^t; i.e., we initialize $F_{p,l} := F_{p2^t,l-1}$ for $p = 0, \ldots, \lfloor \lfloor 2/\varepsilon^2 \rfloor / 2^t \rfloor$, letting $F_{p,l} := \infty$ for $p > \lfloor \lfloor 2/\varepsilon^2 \rfloor / 2^t \rfloor$. After having computed $\underline{U}_l$ and initialized the lth column of F, if object l is small we keep this column unchanged, whereas if object l is large, for $p = \tilde{\pi}_l, \ldots, \lfloor 2/\varepsilon^2 \rfloor$, we let $F_{p,l} := \min\{F_{p,l}, F_{p-\tilde{\pi}_l,l} + w_l\}$.

Extension to the case in which some σ_j is positive is straightforward. Namely, after having computed F and g as above, we consider the $O(|A_i|)$ possible shelf heights, by

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increasing value. For each height h_S, letting l be the object with maximum height among those with height at most h_S, i.e.,

$$l := \arg\max\{h_k\colon k \in \{1, \ldots, s_i\}, h_k \le h_S\}, \tag{23}$$

we consider the lth column of F and complete all the solutions corresponding to the entries whose value is ≤ 1 by either copies of object g_l (as before), or by a fraction of the continuous item of profit $\pi_S := \sum_{j=j_S}^{q_i}(\sigma_j/\sum_{k=1}^{j}\widetilde{w}_k)$, in case this profit is larger than π_{g_l}/w_{g_l}.

Proof of Lemma 3.1. We consider the method illustrated above. It is easy to check that the overall complexity is $O(|A_i|/\varepsilon^2)$. It is also easy to check that $F_{p,l}$ is a lower bound on the minimum capacity required to achieve a profit of $p\delta_l$ by packing large objects among $1, \ldots, l$, after having rounded down the profits of the large objects to the nearest multiple of δ_l. Note in particular that, since the profit of a large object $k < l$ may be initially rounded down to the nearest multiple of $\delta_k = \delta_l/2^t$ for some integer $t \ge 1$, the lower bound may be strict; i.e., we may have a solution of value $p\delta_l$ using capacity $F_{p,l}$ but there may be no such solution after having rounded down all profits to the nearest multiple of δ_l.

What remains to be shown is that the solution produced is near-optimal. First of all, note that each large object has weight $>\varepsilon/2$. Indeed, $\pi_l > \varepsilon \underline{U}_l$ along with $z_l < 2\underline{U}_l$ imply that at most $2/\varepsilon$ copies of object l can be packed in the knapsack. For a given h_S, let z be the maximum profit of a shelf having this height, let z^L and z^S be the contribution of the large and of the small and continuous objects to z, respectively, and let L be the multiset of large objects packed in the corresponding UKP solution. Moreover, let l be defined by (23). The profit of L after rounding down each profit to the nearest multiple of δ_l is

$$\sum_{k\in L} \lfloor \pi_k/\delta_l \rfloor\, \delta_l \ge \sum_{k\in L}(\pi_k - \delta_l) = z^L - |L|\varepsilon^2 \underline{U}_l > z^L - 4\varepsilon z,$$

since $|L| < 2/\varepsilon$ (each large object has weight $>\varepsilon/2$) and $z \ge \underline{U}_l/2$. Accordingly, letting $p := \sum_{k\in L}\lfloor \pi_k/\delta_l \rfloor$, we have $F_{p,l} \le w(L)$; i.e., the object multiset corresponding to $F_{p,l}$ is either L or a multiset with smaller weight. Furthermore, the contribution of the small and continuous items to the optimal solution is

$$z^S \le (1 - w(L)) \cdot \max\{\pi_{g_l}/w_{g_l}, \pi_S\},$$

whereas the additional profit achieved by completing entry $F_{p,l}$ is at least $z^S - \pi_{g_l} \ge z^S - \varepsilon \underline{U}_l \ge z^S - 2\varepsilon z$, since the unused capacity is smaller than w_{g_l} if $\pi_{g_l}/w_{g_l} > \pi_S$, and zero otherwise. Summarizing, we have that the solution obtained by completing $F_{p,l}$ has value at least $(1 - 6\varepsilon)z$, which concludes the proof. □

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